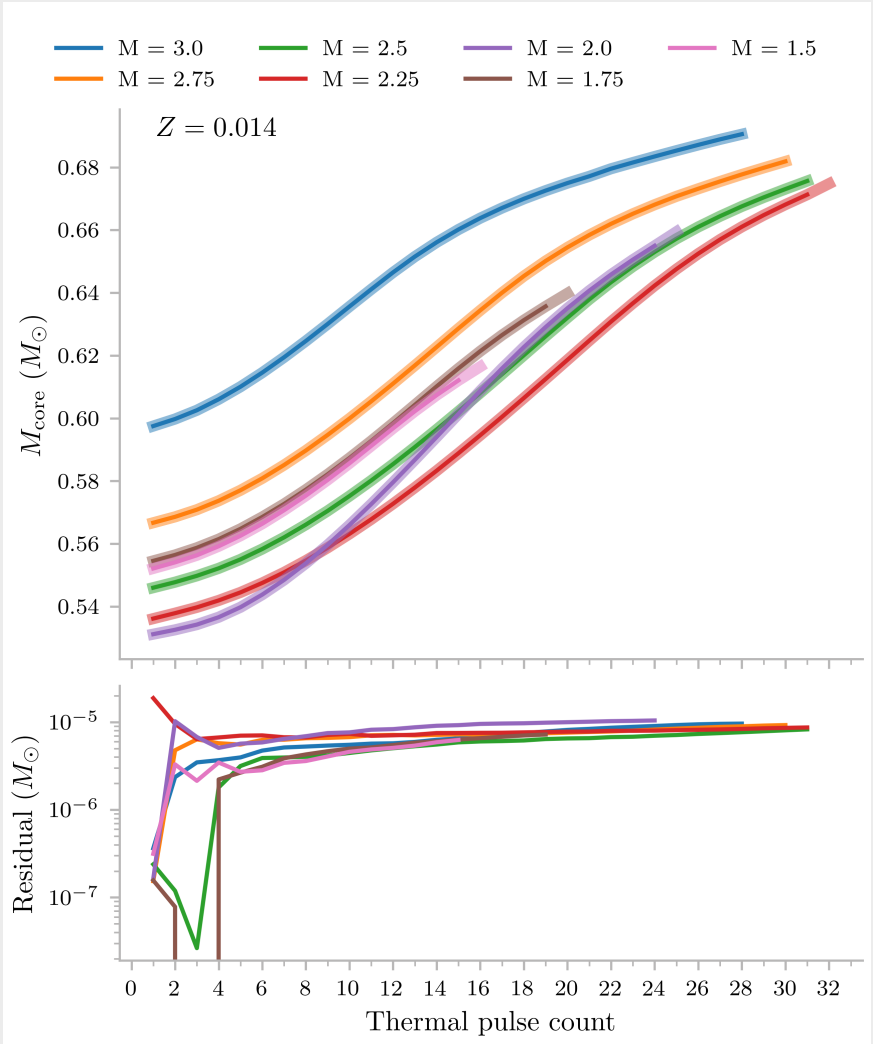


Notes Week 27

a. Correspondence with Amanda Karakas

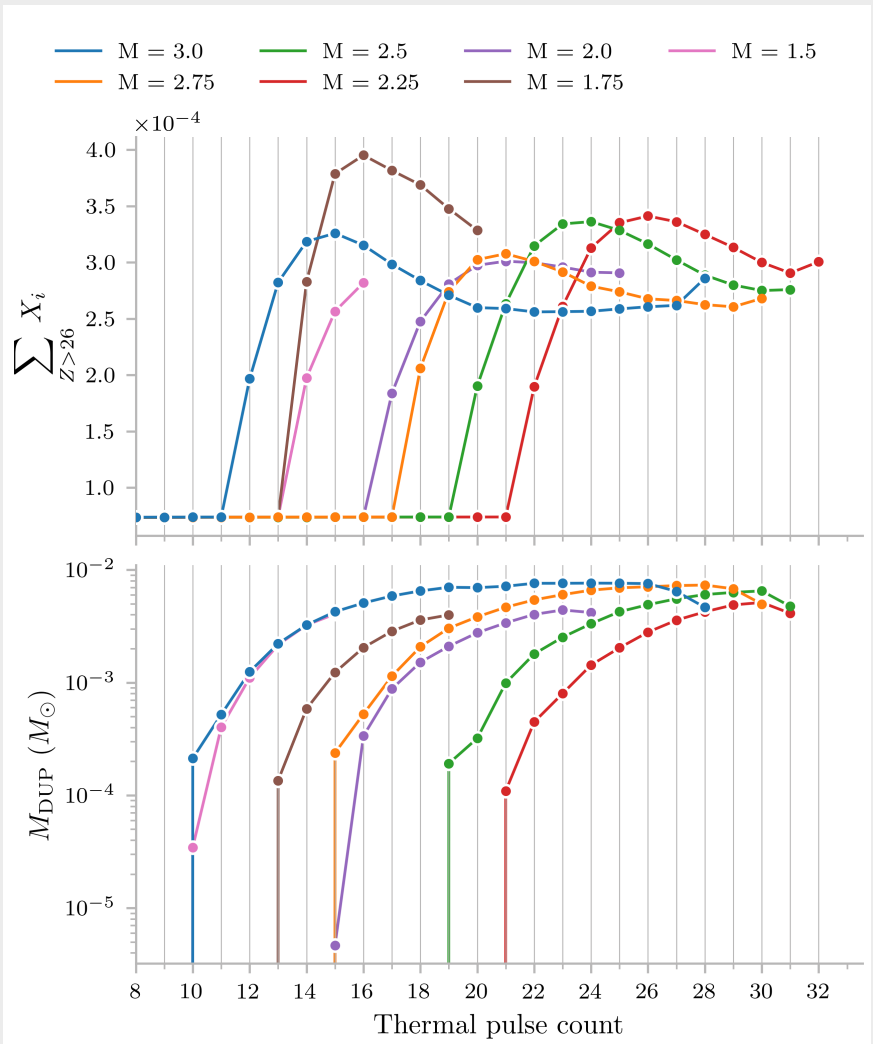
Hi Amanda,
Thanks for your reply and sharing the folder with all of the pulse files!
After looking through the files and discussing with Onno, we do however have one more question that we hope you can help us with. I will briefly describe how we arrived at the question below.
I took some time to match the pulses in the pulse.dat files with the pulses in the intershell abundances files from Zara for all models. This allows me to associate the dredged-up mass from each pulse with the corresponding intershell abundances. To check that this matching is correct, I plotted a quantity that is present in both datasets for every thermal pulse: the core mass.



In the top plot, the thick transparent lines correspond to the core mass per thermal pulse as listed in the intershell files, while the thin opaque lines show the core masses from the pulse.dat files. I think the close agreement between the two shows that the pulses in the two datasets correspond to the same thermal pulses.

This suggests I should be able to combine the dredged-up mass from the pulse.dat files with the corresponding intershell abundances from the intershell files and compare them.

However, when I plot the dredged-up mass per thermal pulse together with the total mass fraction of elements heavier than iron in the intershell, I see something that Onno and I did not expect:



Onno and I expected the intershell to become enriched in s-process elements 1 pulse after the first thermal pulse with significant amounts of dredge up (like what we see for the brown, green, and red models). However, in some models the intershell only becomes enriched after a second TDU (blue and purple models) or even after a third or fourth TDU (pink and orange models). We see similar offsets in the models with different metallicities as well.

Do you happen to know what might cause some models to only experience s-process enrichment in the intershell after multiple TDU events?

Thanks again for your help!

Best regards,

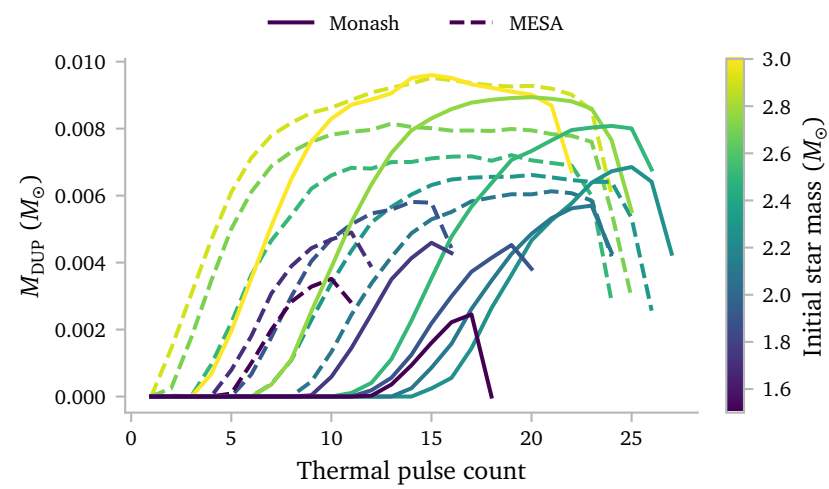
Koen

”

b. Comparing MESA and Monash Dredge up

My first idea estimate the intershell abundances of the MESA models by parametrising the intershell abundance as a function of cumulative dredge up. This makes little physical sense, but is convenient because the cumulative amount of dredge up is strictly increasing.

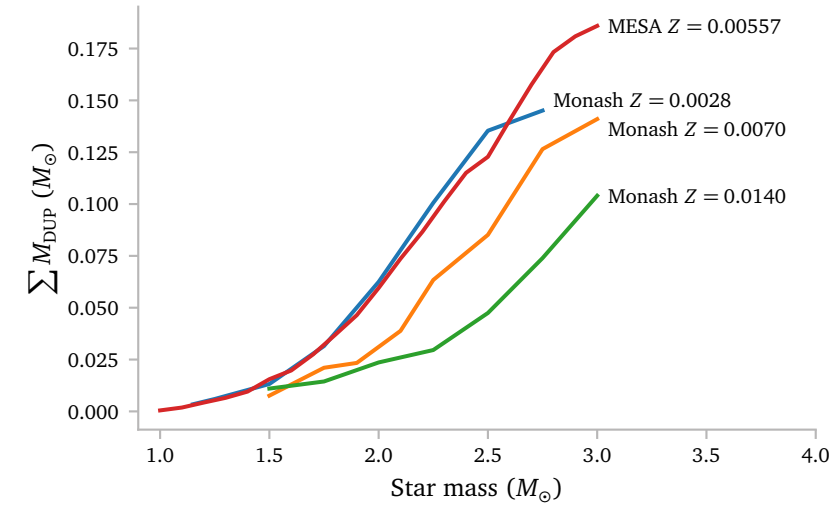
However, this is only a good way to approximate the intershell abundances if the dredge up episodes of the MESA models are in good agreement with the dredge up episodes of the Monash models.



It is important to note that I plot the Monash models with $Z = 0.007$ and MESA models with $Z = 0.00557$, so there is a slight difference there.

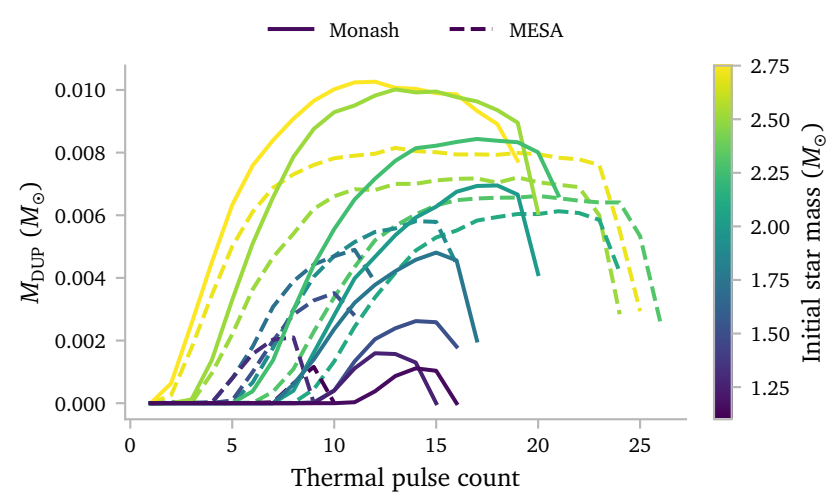
It seems like the MESA models generally undergo dredge up earlier. The magnitudes are quite similar, but because the MESA models start dredge up earlier, I think they will in general experience a bit more total dredge up

Below, I plot the total amount of dredge up for the various Monash and MESA models.



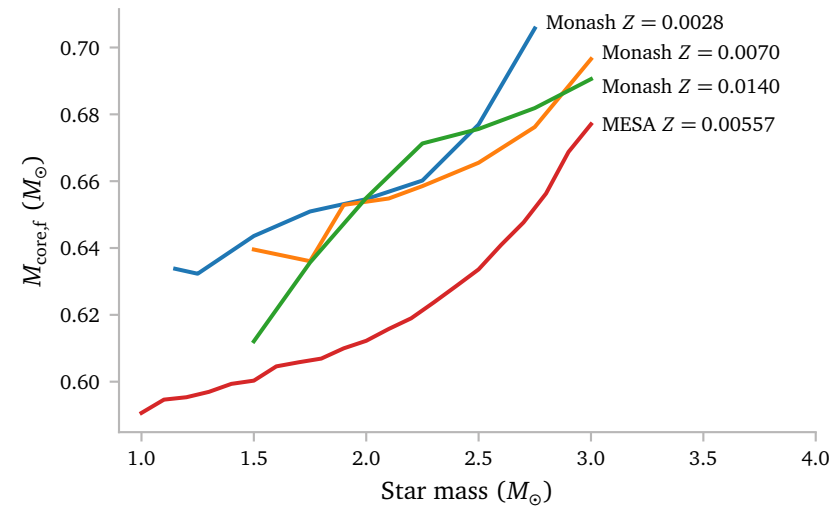
The MESA models agree very well with the lower $Z = 0.0028$ metallicity Monash models. Obviously, since these are entirely different codes, it is to be expected that they do not predict the same amount of dredge up, but I feel like the difference here and in the previous plot is actually quite significant, especially if I use the cumulative dredged up mass to predict the intershell abundance.

Since the Monash $Z = 0.0028$ models seem to agree very well with the MESA models when it comes to total dredged up mass, I want to compare them pulse by pulse as well.



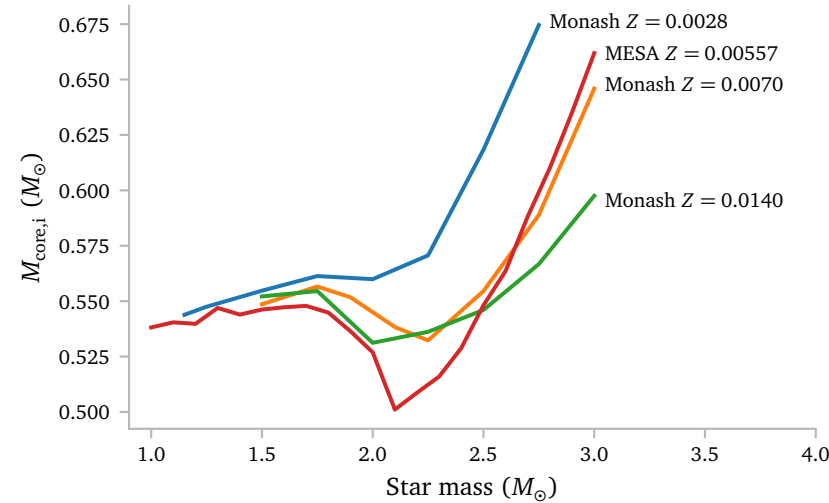
It seems like it is a coincidence that the cumulative amount of dredged up mass agrees so well. The MESA models generally experience more thermal pulses with dredge up, while the amount of dredged up mass per pulse is lower.

Apart from the amount of dredged up mass, there are some other properties that we can compare between the models. First, let me show the final core mass as a function of stellar mass:

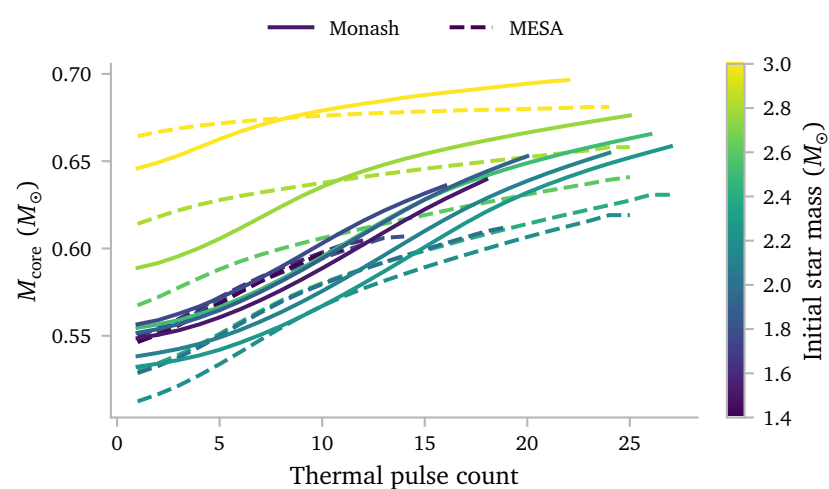


The final core mass of the MESA models is always lower than that of the Monash models. It think it is not super surprising given the fact that we saw that the MESA models experience *more* dredge up, and thus grow their core *slower*.

I think it is interesting to look at the initial core mass too though, because this is a good check if the pre-TPAGB evolution could be a cause for this difference too.



The initial core mass is also not identical, but at least for the higher masses, MESA and Monash agree well. I do think it is interesting that the core mass of the MESA models is much lower for the models that just miss the core Helium flash.

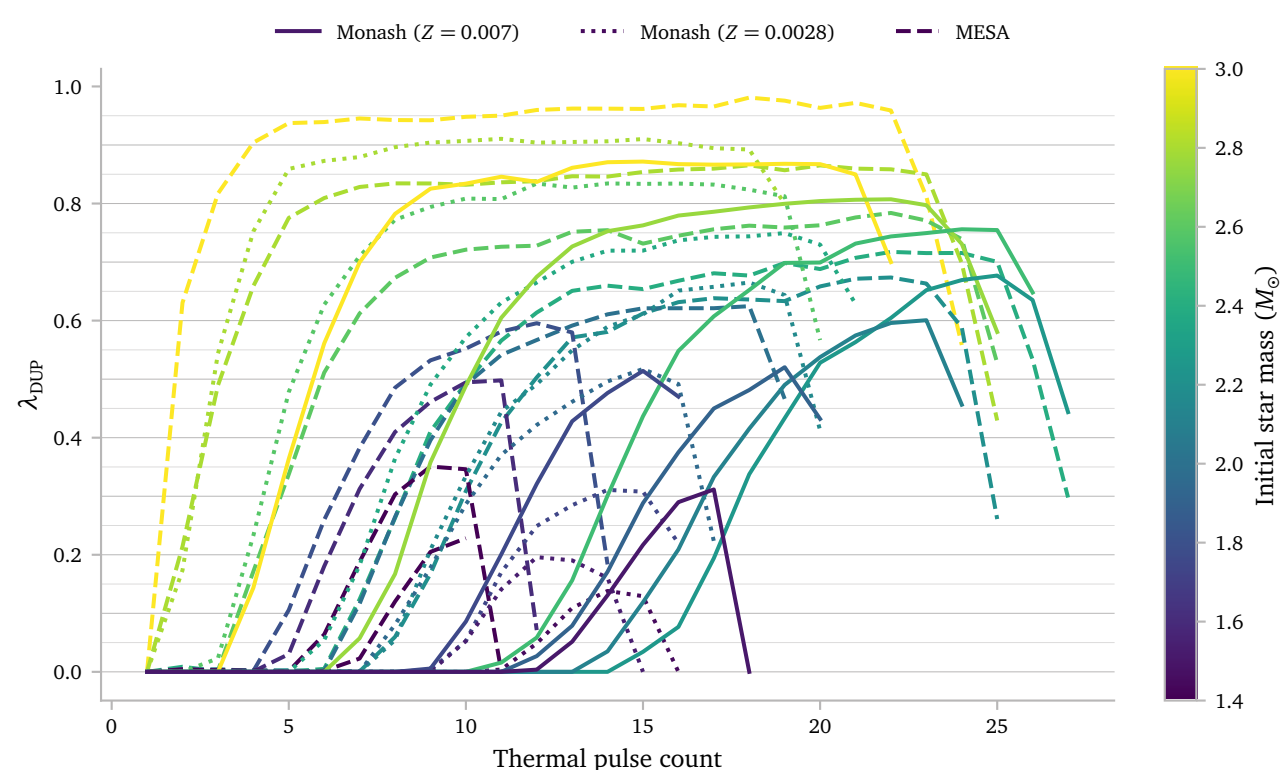


The core mass evolution of the MESA models is *much* more flat than that of the Monash models. This is exactly what we expect based on the lower total dredged up mass.

I do think it is a bit surprising that we find such large differences especially in the core mass, since Cinquegrana et al. (2022) found only very small differences in the core mass of a $2 M_\odot$ $Z = 0.03$ star. However, they did specifically try to match every parameter of Monash to the equivalent MESA parameter, which we did not do.

Rees et al. (2024) also looked at the core masses of MESA models and Monash models. Just like us, they also find that for low mass TPAGB stars, the MESA core masses are lower than those of the Monash models. They do note that MESA and Monash have slightly different definitions of the core mass, however, in both cases it is based on the hydrogen burning shell. They note that, since the hydrogen burning shell is very thin in AGB stars, the differences in the core masses are *not* due to different definitions.

The plot below shows the dredge up efficiency parameter λ for the MESA models, as well as for the $Z = 0.0028$ and $Z = 0.007$ Monash models.



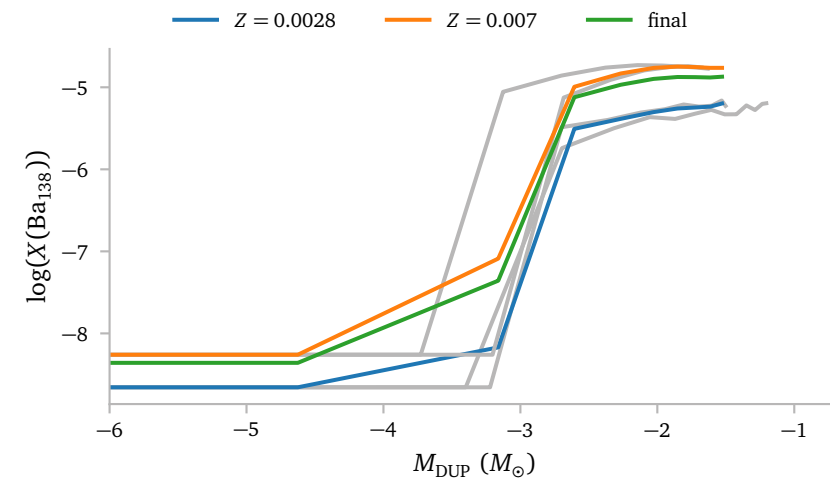
The shapes here are actually quite similar for all models. It is just that the MESA models seem to predict higher overall values.

c. Yields

d. Comparing different Intershell abundance parametrisations

d.1 Summary of Different Methods

1. *Dredged-up mass*: This method parametrises the isotope intershell abundance only based on the total amount of dredged-up mass. This is nice because, just like for example the thermal pulse count, this quantity is strictly increasing, and it has some physical meaning since we expect the intershell abundance enrichment to only start increasing after a dredge-up has started. However, it makes little physical sense to say that the intershell abundance is dependent on the *cumulative* amount of dredged up mass. Really, the only important thing is the *first* time significant dredge up occurs.



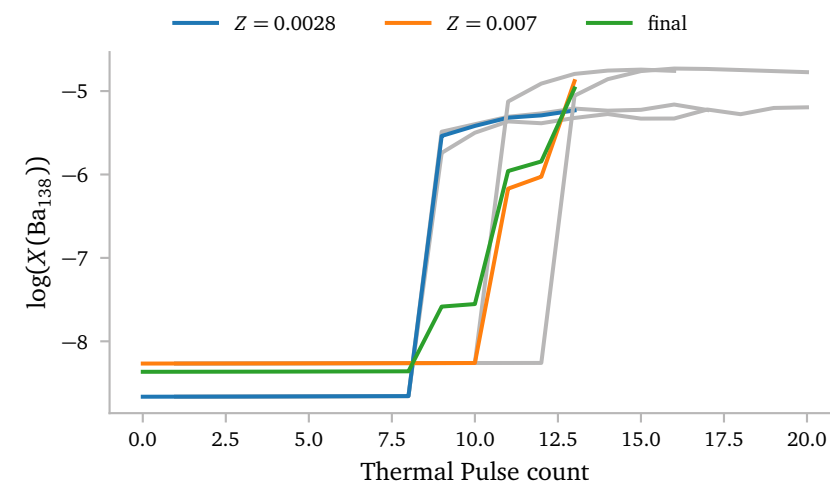
We see that this method is quite good at interpolating the latter part of the isotope abundance evolution as a function of cumulative dredged-up mass. However, there is not a very well defined “start” of the intershell abundances *because* the abundances increase very quickly after the first or second dredge-up event. These events can dredge up quite different amounts of mass.

2. *Naive thermal pulse*: This method only looks at the intershell abundance of an isotope as a function of the thermal pulse number. We then *assume* that the MESA and Monash models agree well on when the intershell abundances change as a function of thermal pulse number.

This is probably *not* the case, but it is an interesting simple test case. I assume that this method is not very accurate because the MESA and Monash models do not agree on the total number of thermal pulses, and they do not agree on when the first pulse with significant dredge up occurs.

Another reason why I think this method is not ideal is that we *need* to interpolate intershell abundances of the different monash models, and even these models do not agree well on during which thermal pulse enrichment occurs. This means that interpolation will give “smoothed-out” results.

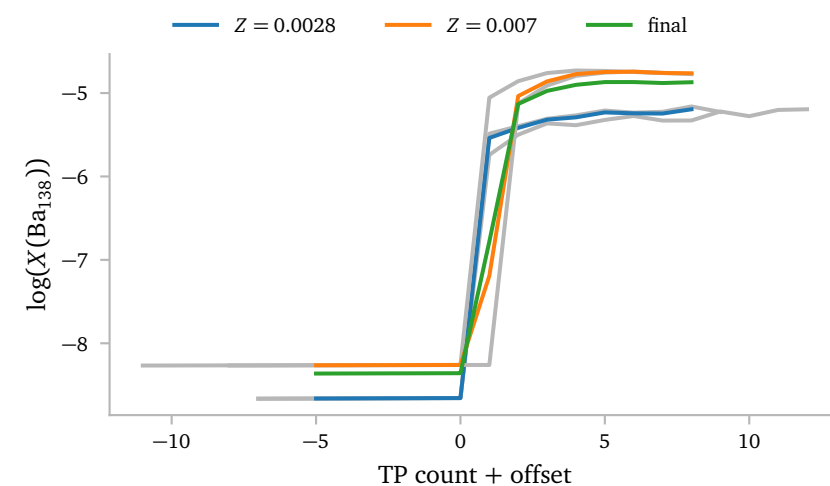
Below I show how a $M = 1.8 M_{\odot}, R = 500 R_{\odot}, q = 0.4293$ models gets interpolated:



Here we see the “problem” with this method. Since for the same metallicity, the different masses do not agree on the moment of enrichment, the first interpolation creates a smoothed intershell abundance enrichment profile for the $Z = 0.007$ models. Then when we interpolate the different interpolated Z models, we get an even more smoothed out profile. This is *not* representative of how the single model intershell enrichment looks.

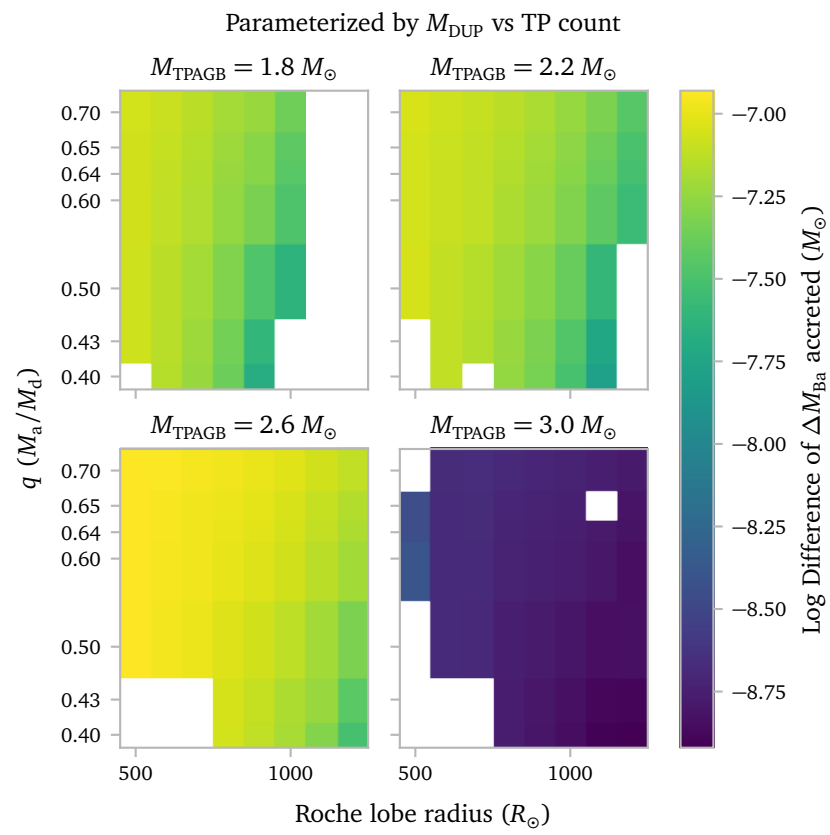
3. *Thermal pulse shifted by first significant dredge-up*: Both of the previously noted methods clearly have problems. We can try to take the best parts of both. Let’s take the simplicity of the thermal pulse count idea and the physical meaning of the first dredge-up that is so powerful in the first method.

For now I put a simple lower limit of $\log M_{\text{DUP}} = -4.5 M_{\odot}$ for the start of dredge up. I then offset the thermal pulse count by the first pulse that satisfies the lower boundary.



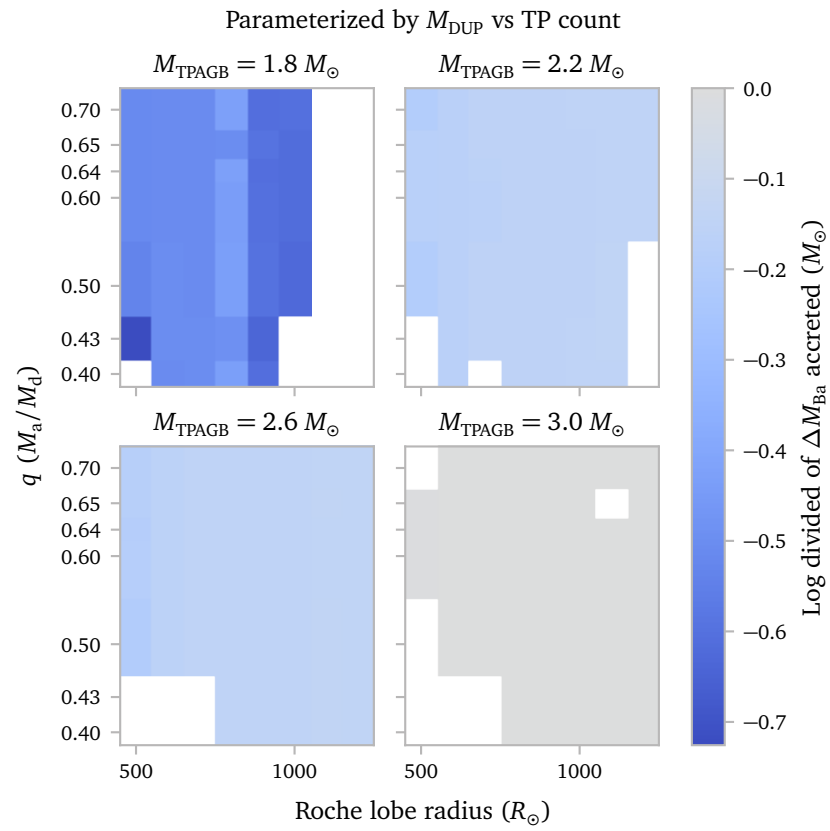
I think this looks quite good. The models however do not perfectly agree on when the intershell should become enriched. We have already seen this before and I have asked Amanda about this, but at this point I have not gotten a response yet.

d.2 Comparing Naive Thermal Pulse parametrisation with Cumulative Dredged-up Mass parametrisation



It is clear that the absolute differences between the two methods for determining the intershell abundance, and thus the total amount of accreted barium is smallest for the $M_{\text{TPAGB},i} = 3 M_{\odot}$ model and largest for the two lightest models. However, we do not know what the absolute values of ΔM_{Ba} are. Therefore, I also plot the log-divided difference between the methods.

Here, I plot $\log(M_{\text{dup method}} - \text{TP count method})$

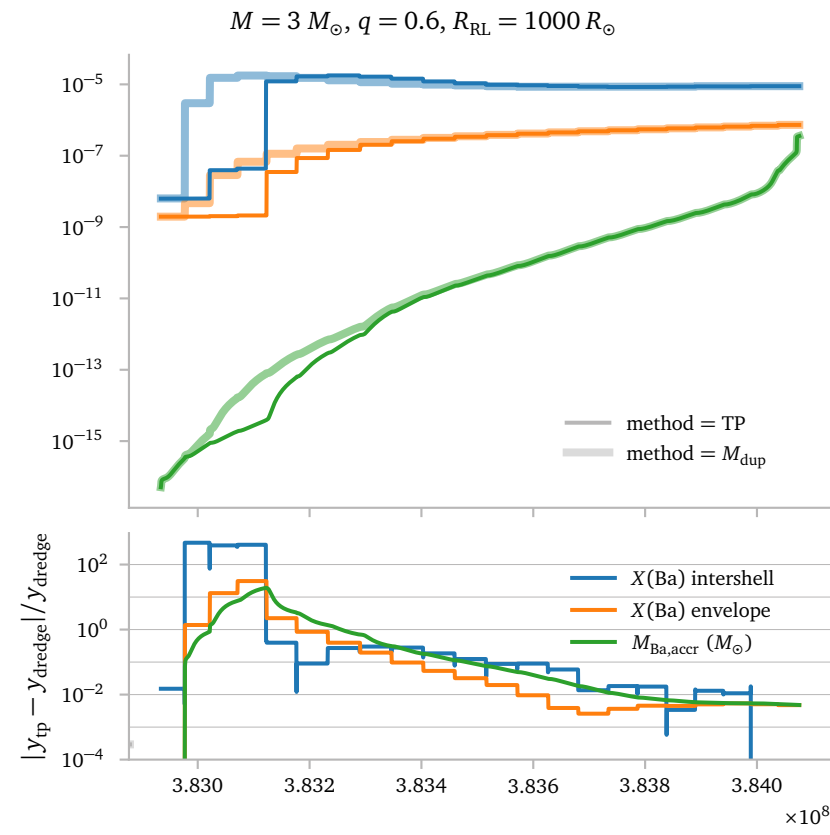


This plot shows $\log(M_{\text{dup method}}/\text{TP count method})$

This time it is more obvious that the differences are tiny for the $M = 3 M_{\odot}$ models and largest for the $M = 1.8 M_{\odot}$ models.

Now the question is, *why* do these differences exist?

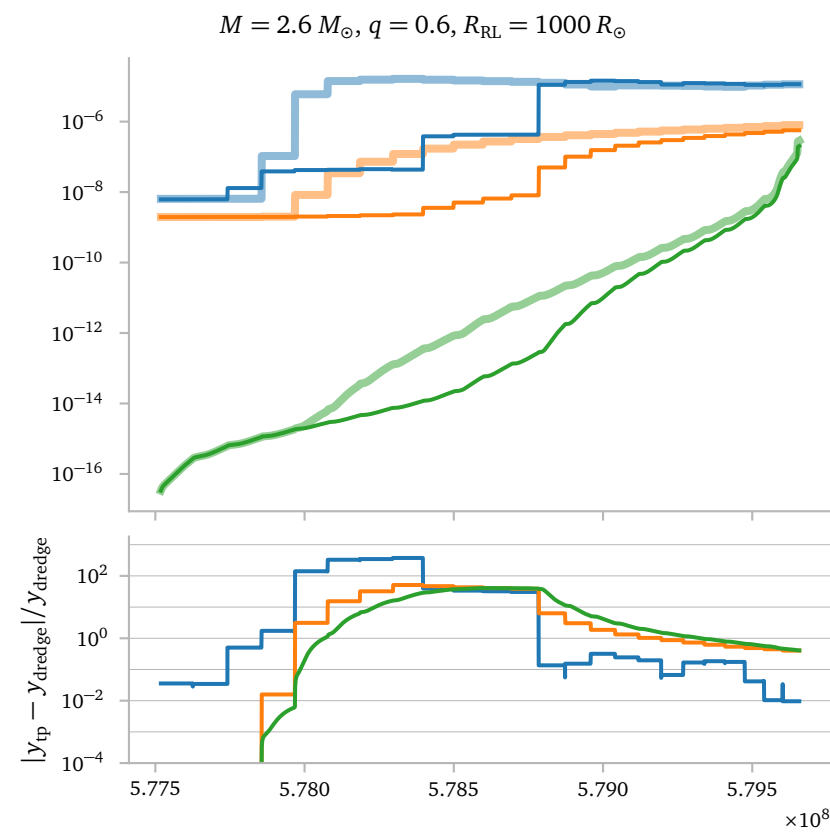
First, let me show a 3 $M_{\odot}$ model.



Now as we could see in the previous figure, the $M = 3 M_{\odot}$ models have very little differences between the amount of accreted barium mass.

I think this is because in these models, dredge up starts early in both the MESA and the Monash models, such that overall, the differences in the *envelope* abundances remain small after some dredge up events. This also means that as soon as the star starts to lose significant amounts of mass, well after dredge up has begun, the envelope abundances agree well and we are left with roughly the same amount of accreted barium.

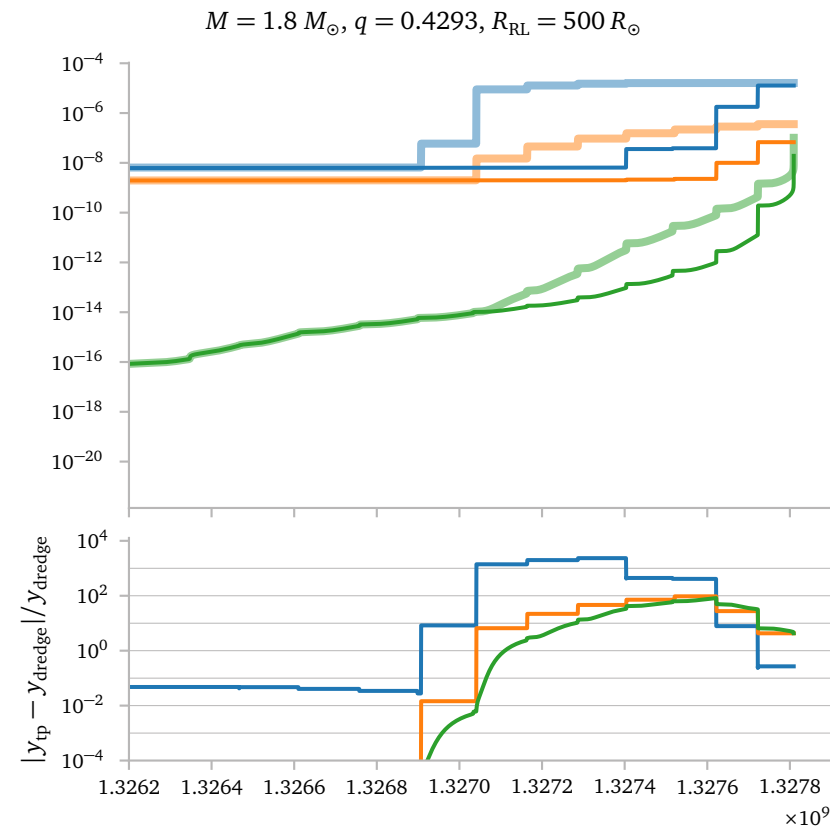
Now this is not only true for barium, but pretty much all s-process elements, because the enrichment happens in the same way.



In the differences plot above, we see that the $M = 2.6 M_{\odot}$ models generally have lower values by about 1.5 to 2.

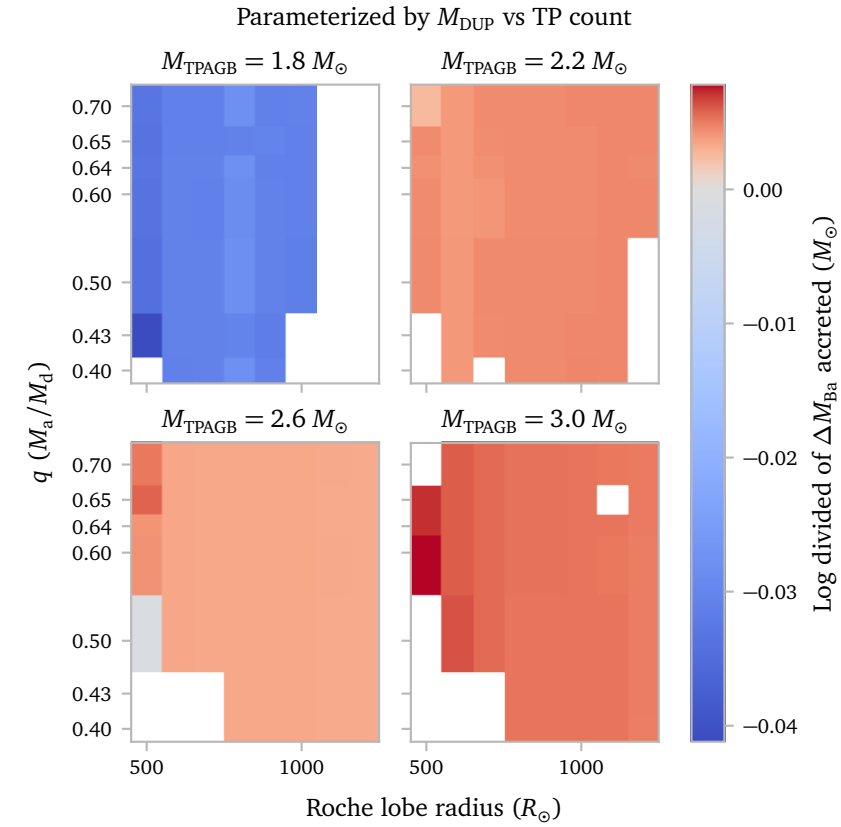
Here, we see that it takes a lot longer for the intershell to become enriched. It also seems like it takes many pulses for the intershell to actually reach its final intershell abundance maximum, while the Monash models reach this maximum in only 1 or 2 pulses. This is a result of the interpolation which is definitely not perfect.

Luckily, there are *many* pulses after the first TDU event. This way, even though the envelope is initially much less enriched, it can “catch” up before significant amounts of mass are transferred. Once the strong winds and RLOF start, the envelope is nearly at the same enrichment as it is for the M_{DUP} interpolation method.



This model has the largest fractional difference between the methods. It is quite large, by this is 2.6 $M_{\odot}$ model, the difference between when the intershell becomes enriched is quite large. This means the intershell becomes enriched much earlier in the M_{DUP} method than it does in the TP count method. However, because this model loses most of its mass very quickly after the intershell enrichment of the TP count method, it does not have any time to properly enrich its envelope, while the M_{DUP} method did have time to enrich it.

d.3 Comparing Thermal Pulse Offset parametrisation with Cumulative Dredged-up Mass parametrisation



References

Cinquegrana, G. C., Joyce, M., & Karakas, A. I. (2022) *Bridging the Gap between Intermediate and Massive Stars. I. Validation of MESA against the State-of-the-Art Monash Stellar Evolution Program for a $2 M_{\odot}$ AGB Star.* [ApJ939](#), 50.

Rees, N. R., Izzard, R. G., & Karakas, A. I. (2024) *Thermal pulses with MESA: resolving the third dredge-up.* [MNRAS527](#), 9643.

A Optimizations

Last week, my code to compute the intershell abundances was very unoptimized. It worked but very often computations were repeated when it was not necessary.